

Learning-Augmented Influence Maximization with Audited Marginal-Gain Oracles

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Abstract

Influence maximization selects a small set of seed nodes to maximize information diffusion in a network, but high-quality sequential selection requires repeatedly estimating conditional marginal gains with expensive influence oracles. Learned predictors can provide fast marginal-gain estimates, yet treating them as replacements for classical oracles is brittle: prediction errors are state dependent and can accumulate over the seed-selection trajectory. We study a learning-augmented alternative in which learned marginal gains are used as fallible advice rather than trusted decisions. Our framework first performs cheap full-graph screening, then combines a state-aware marginal proposal model with residual auditing, progressive verification, and a classical fallback. The resulting solver adaptively spends less oracle computation when learned advice is informative and automatically increases verification when the predictor becomes unreliable. Preliminary full-graph experiments support a favorable quality–efficiency tradeoff while controlled predictor corruption triggers the intended fallback behavior. These results suggest that reliable learning for influence maximization should focus not only on prediction accuracy, but on when and how predictions are verified inside the sequential optimization loop.

1 Introduction

Influence maximization (IM) is a canonical sequential optimization problem on social and information networks: given a diffusion graph and a seed budget, the goal is to select a small set of nodes whose activation maximizes expected influence spread. The classical greedy view is conceptually simple, but each greedy step can require comparing conditional marginal gains for many candidates, and accurately estimating these gains through Monte Carlo simulation or scalable sampling-based influence oracles remains computationally expensive on large graphs.

Learning offers an appealing shortcut. A graph model can predict which candidates are promising, or directly estimate the marginal gain of adding a candidate under the current seed set. However, a sequential IM solver needs more than an accurate predictor on average. The relevant ranking changes after every selected seed because of overlap and diminishing returns; a candidate that looks strong globally can become redundant under the current state. Consequently, a locally plausible prediction error can remove the true best candidate from consideration and propagate through later decisions. This makes a fully learned replacement for the influence oracle difficult to trust under distribution shift or predictor failure.

We take a different perspective: a learned marginal predictor should be treated as fallible advice to a classical optimizer, not as an oracle that must always be correct. The key question is therefore not “Can a GNN replace influence estimation?”, but rather “When is learned advice useful enough to reduce oracle work, and how much verification is needed before acting on it?” We instantiate this perspective with a learning-augmented IM framework that combines full-graph candidate screening, state-aware marginal proposals, an audited residual trust test, progressive candidate and sample verification, and a classical fallback. When predictions are useful, the learned model concentrates expensive evaluation on a small

head of plausible candidates. When predictions are unreliable, the audit expands real-oracle computation and can recover classical behavior.

Our current evidence supports three central claims. First, marginal prediction must be explicitly state aware: supervision that exposes the same candidate under different seed states substantially improves conditional ranking sensitivity. Second, prediction alone is insufficient for reliable seed selection, whereas selective verification recovers much of the lost decision quality. Third, an audited progressive solver yields the desired learning-augmented behavior: useful advice reduces expensive verification, while corrupted or random advice increases oracle effort and activates fallback rather than causing catastrophic downstream degradation. Importantly, this behavior persists when the method receives the full influence graph and uses an independent screening stage to construct a compact candidate set.

Contributions. Our contributions are threefold:

- We formulate learned marginal-gain prediction for sequential IM as a fallible advice interface, highlighting why ordinary offline prediction quality is insufficient to guarantee reliable downstream decisions.
- We propose a learning-augmented solver that integrates full-graph screening, state-aware marginal proposals, residual auditing, progressive verification, and fallback to classical influence evaluation.
- We evaluate the resulting quality–efficiency–robustness tradeoff, including controlled predictor degradation and full-graph inputs, and show that oracle computation adapts to advice quality rather than assuming a uniformly reliable learned model.

2 Related Work

Classical influence maximization and scalable influence estimation. We will position the method against greedy influence maximization, Monte Carlo evaluation, and modern reverse-influence-sampling (RIS) solvers. The final version will emphasize strong paper-grade RIS baselines rather than weak heuristics.

Learning-based influence maximization. Prior GNN- and reinforcement-learning-based approaches learn node scores, seed-selection policies, or value/marginal surrogates. Our distinction is not that marginal gains have never been predicted; it is the explicit online interface that audits learned advice, allocates classical verification adaptively, and falls back when advice is unreliable.

Learning-augmented algorithms and prediction with verification. Our framework is most closely related in spirit to learning-augmented optimization: predictions are used to improve efficiency when helpful while a non-learned mechanism limits damage from poor advice. The final literature review will sharpen the connection to consistency/robustness and safe-verification principles without overstating guarantees that are not proved here.

3 Learning-Augmented Influence Maximization

3.1 Problem and Learned-Advice Interface

Let $G = (V, E)$ denote a diffusion graph and let $\sigma(S)$ be the expected spread of seed set $S \subseteq V$. Given a seed budget k , influence maximization seeks

$$\max_{S \subseteq V, |S| \leq k} \sigma(S). \quad (1)$$

At a sequential state S , the conditional marginal gain of candidate $v \notin S$ is

$$\Delta(v \mid S) = \sigma(S \cup \{v\}) - \sigma(S). \quad (2)$$

We assume access to a learned advice model $\hat{\Delta}(v | S)$ that is inexpensive relative to a high-fidelity classical influence oracle. Crucially, we make no assumption that the advice is uniformly accurate.

3.2 Full-Graph Candidate Screening

The full graph is first reduced to a high-recall shortlist C_M using a cheap global RR/RIS sketch. This separates global scalability from the harder sequential conditional-ranking problem. Unless otherwise specified, our default shortlist size is $M = 128$, while $M \in \{64, 128, 256\}$ is used to expose the quality–cost sensitivity.

3.3 State-Aware Marginal Proposal

For candidates in $C_M \setminus S$, a graph predictor estimates $\hat{\Delta}(v | S)$. Training explicitly exposes candidate nodes under multiple seed states and supervises conditional changes in gain, reducing the candidate-strength shortcut in which a model predicts mostly static node quality while ignoring diminishing returns.

3.4 Audited Residual Trust Test

At each decision step, we evaluate a small predicted head together with rank-spaced sentinel candidates using a limited-budget classical oracle. For each audited candidate, define the residual

$$r(v | S) = \Delta_{\text{audit}}(v | S) - \hat{\Delta}(v | S). \quad (3)$$

The audited residuals are used to estimate whether an unverified outsider can plausibly overturn the current audited winner. Unlike a generic correlation statistic, this trust test is aligned with the downstream failure event we care about: excluding a candidate that could beat the verified head.

3.5 Progressive Verification and Fallback

If the audit indicates that the current head is safe, we progressively increase candidate coverage and influence-estimation budget only as needed. If the residual audit signals unreliable advice, the solver expands verification and, in the limiting case, falls back to all-candidate classical evaluation. Thus the learned model proposes where to spend oracle computation, while classical verification retains control of the final sequential decision.

4 Analysis

The theoretical goal of this paper is deliberately modest and aligned with what the algorithm actually guarantees. We will formalize (i) advice error, (ii) audit coverage, and (iii) verification cost. A conditional correctness proposition will show that, whenever the audited upper envelope validly covers the residual of every unverified outsider, the selected candidate matches the decision that would be obtained by the corresponding classical oracle. This yields a deterministic efficiency envelope: informative advice can reduce the number of expensive evaluations, whereas untrusted advice increases verification and the worst case recovers classical computation. We do not claim an unconditional approximation guarantee from the learned predictor itself.

5 Experiments

We organize the experiments around five research questions rather than around implementation modules.

RQ1: Is learned marginal gain genuinely state aware? We evaluate cross-state conditional ranking, same-candidate sensitivity, and diminishing-return response under strict unseen-state and overlap-stress settings.

RQ2: Can learned advice reduce oracle work without sacrificing influence quality? We compare learned-only selection, fixed verification, the proposed audited progressive solver, and strong classical references using influence quality, oracle work, and wall-clock/runtime breakdowns.

RQ3: Does the solver degrade gracefully when predictions become unreliable? We inject controlled corruption from clean through severe and random predictors. A successful learning-augmented method should increase verification cost as advice worsens while keeping final spread close to a classical reference.

RQ4: Does the method remain valid from full-graph inputs? We run independent full-graph RR/RIS screening before the learned sequential stage and report shortlist recall together with end-to-end quality and cost.

RQ5: Is the tradeoff stable across shortlist sizes, seed budgets, and graphs? We test $M \in \{64, 128, 256\}$, multiple seed budgets, and multiple standard IM graphs. Paper-grade RIS baselines and final ablations are added here as the remaining experiments complete.

Current preliminary evidence. On NetHEPT full-graph runs, the current closure experiments retain strong seeds in the screened shortlist and achieve approximately 98.9% of the RR-reference influence quality while using roughly 59.1% of the expensive Monte Carlo verification cost. These numbers are preliminary and will be replaced by the final multi-dataset, multi-budget table before submission.

Reproducibility Statement

We will provide the complete algorithmic specification, training protocol, dataset preprocessing, influence-model parameters, evaluation budgets, random-seed protocol, and all ablation settings in the main paper and appendix. Anonymous code and configuration files will be included as supplementary material so that the reported screening, marginal-prediction, verification, corruption, and fallback experiments can be reproduced from the same pipeline.

AI Use Statement

Generative AI tools were used during this project to assist with research planning and feedback on experimental design, software-code editing, literature search and summarization, interpretation and organization of experimental results, and drafting and editing parts of the manuscript. AI-assisted outputs are reviewed against source materials, executable experiments, and the authors' own analysis before inclusion. The authors take responsibility for the final content of the paper, including all claims, citations, code, experiments, and text produced with the assistance of generative AI. This disclosure is a working draft and will be reconciled with the complete research workflow before submission.

References

Bibliography placeholder for Draft v0.1. References will be populated from the verified GRL literature library.

A Additional Experimental Details

This appendix will contain full hyperparameters, dataset statistics, detailed corruption protocols, additional M and k sensitivity, per-seed results, runtime decomposition, and extended ablations.

B Additional Analysis

Complete proofs and any finite-sample audit discussion will be placed here once the corresponding statements are frozen.